INCH-POUND
MS24149G
w/AMENDMENT 1
10 October 2024
SUPERSEDING
MS24149G
15 December 2021

MILITARY SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 2 PDT, 10 AMPERES, TYPE I, HERMETICALLY SEALED

Reactivated after 15 December 2021 and may be used for new and existing design acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

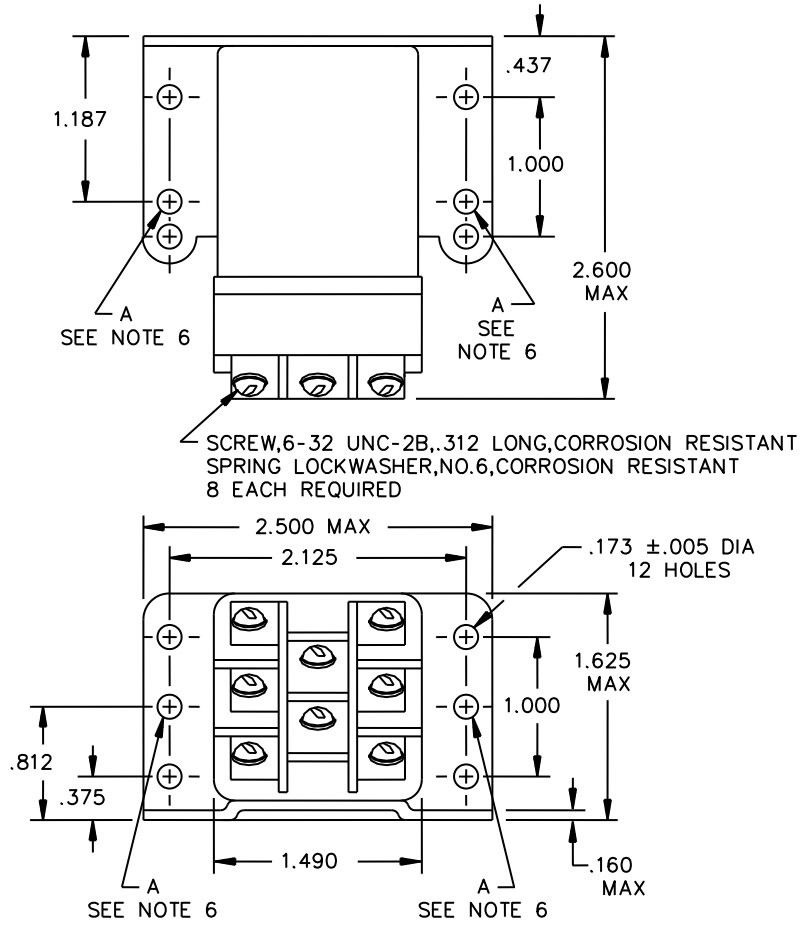


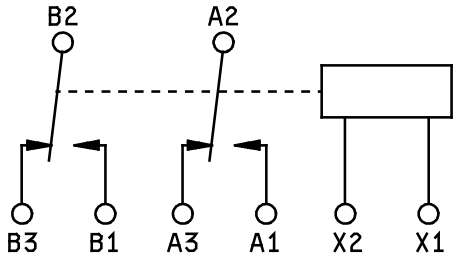

FIGURE 1. Dimensions and configurations.

AMSC N/A

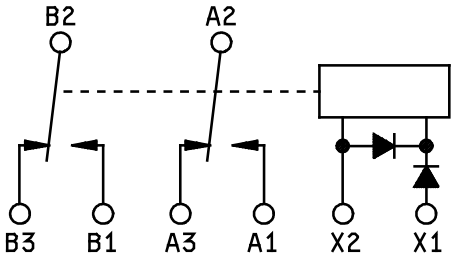
FSC 5945

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CIRCUIT DIAGRAM
DC COILS




CIRCUIT DIAGRAM
AC COILS
SEE NOTES 7 AND 8

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Part number MS24149-D1 replaces part numbers AN3311-1 and AN3311-2.
5. Interchangeability relationship - MS24149 parts can universally replace the cancelled AN3311 parts but cannot always replace the superseding MS24149 parts.
6. Mounting holes 'A' to be used only when replacing AN3311.
7. The use of diodes on ac relays is optional. Actual application must be shown on label.
8. Ac coils are inactive for new design.
9. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.
10. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this specification sheet to the extent specified herein.
11. Cadmium or cadmium compounds are prohibited on external hardware. A transition period to non-cadmium hardware is authorized for up to 1 year from the date of this revision.

FIGURE 1. Dimensions and configurations - Continued.

TABLE I. Dash numbers and characteristics.

| Dash number MS24149- | Type | Coil | Terminal type | Mounting or mating socket | Auxiliary contacts | Max weight in pounds |
|---|---|---|---|---|---|---|
| D1 | I | dc | Screw | Bracket | N/A | 0.44 |
| A1 1/ | I | ac | Screw | Bracket | N/A | 0.44 |

1/ Dash number -A1 is inactive for new design and shall be used for support of existing equipment designs only.



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TABLE II. Operating characteristics.

| PIN MS 24149- | Coil data | | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Drop out volt-age 2/ | Hold volt-age 2/ | Oper-ate 3/ | Re-lease 4/ | Contact bounce | | | |
| | | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res +15% -10 | Volts | Amp | Nor-mal 2/ | High temp test | Cont cur-rent test | | | | | NO | NC | NO | NC |
| D1 | dc | 28 | N/A | 160 | 29 | 0.25 | 18 | 19.5 | 22.5 | 1.5 | 7.0 | 20 | 20 | 2 | 2 | --- | --- |
| A1 | ac | 115 | 400 | N/A | 120 | 0.06 | 90 | 95 | 103 | 5 | 30 | 20 | 50 | 2 | 2 | --- | --- |

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) (case grounded).

| Type of load | Life operat-ing cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | | See appropriate notes |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | |
| Resistive | 50 | 10 | 10 | | | 10 | | | | | | | | |
| Inductive | 10 | 10 | 10 | | | 10 | | | | | | | | |
| Inductive | | | | | | | | | | | | | | |
| Motor | 50 | 6 | 6 | | | 6 | | | | | | | | |
| Lamp | 50 | 2 | 2 | | | 2 | | | | | | | | |
| Transfer load | | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 200 | 2.5 | 2.5 | | | | | | | | | | | |
| Mixed loads | 50 | 0.100 | 0.100 | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.

Environmental characteristics:

| | |
|---|---|
| Temperature range | -70°C to +125°C. |
| Max altitude rating | 80,000 ft. |
| Shock G-level | 25 G. |
| Duration | 11 ms. |
| Max duration contact opening | 10 μs. |



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Vibration – sinusoidal,

G-level 10 G.

Frequency range 10 - 1,500 Hz.

Acceleration 15 G.

Electrical characteristics:

Minimum insulation resistance, initial 100 megohms.

After life or environmental tests 50 megohms.

Dielectric strength (sea level).

| | Initial | After life tests |
|---|---|---|
| Coil to case | 1,250 V rms | 1,050 |
| Aux contacts | --- | --- |
| All other points | 1,500 V rms | 1,250 |

Dielectric strength (altitude): 1 minute.

| | Initial | After life tests |
|---|---|---|
| Coil to case | 500 V rms | 500 V rms |
| Aux contacts | 500 V rms | --- |
| All other points | 500 V rms | 700 V rms |

Max contact drop initial: 0.150 volt.

After life test: 0.175 volt.

Overload current (NO): 40 amperes dc, 40 amperes ac.

Rupture current 50 amperes dc, 50 amperes ac.

Duty rating: Continuous.

RFI specification: MIL-STD-461.

(Applicable to coil circuits of ac operated relays).

Conformance inspection:

Group A acceptance reports shall be submitted to the qualifying activity every 36 months in order to retain qualification for this military specification sheet.

Performance of groups B and C tests may be suspended at the discretion of the qualifying activity.

Qualification by similarity: See MIL-PRF-6106.



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Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations."
requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-6106, this document references the following:
MIL-STD-461

Custodians:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2024-057)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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